

A Hybrid Deep Learning Framework for Multimodal Biometric Verification Utilizing ECG, Fingerprint, and Iris Data

Dr. T. Raghunadha Reddy¹, B. Naga Mallesh², K.V. Sai Jayanth³, and D. Raju⁴

¹Associate Professor & HOD, ^{2,3,4}Student

Department of Computer Science, Matrusri Engineering College, Hyderabad, India

Email: {drtrnreddy, cse22733129, cse22733147, cse22733187}@matrusri.edu.in

Abstract—Single-trait biometric systems frequently face critical security flaws, ranging from physical sensor wear to sophisticated spoofing attacks. When deploying modern deep learning to solve these issues, developers encounter a new problem: data starvation. Standard classification models require thousands of images to generalize, leading to massive overfitting when only a few enrollment scans are available. To overcome this limitation, we engineered a robust multimodal biometric architecture tailored specifically for data-scarce constraints (a maximum of three training samples per user). Our solution fuses a dynamic, one-dimensional behavioral signal (Electrocardiogram) with static, two-dimensional physical traits (Fingerprint and Iris) across a dataset of 199 unique individuals. To bypass the failure of standard Convolutional Neural Networks (CNNs) on sparse physical data, we shifted the network objective from 1-to-N identification to 1-to-1 verification using Siamese architectures. We processed the physical traits through parallel Siamese ResNet50V2 networks optimized with contrastive loss to generate 512-dimensional feature embeddings. Simultaneously, a custom-built 1D-CNN evaluated the temporal ECG data. We then normalized and merged the outputs from these varying modalities using a targeted, weighted score-level fusion algorithm. Testing showed that while traditional 2D Softmax classifiers stalled below 40% accuracy, our Siamese metric learning approach drastically elevated unimodal performance. The combined multimodal system (anchored by a 50% ECG, 30% Iris, and 20% Fingerprint weight distribution) achieved a final accuracy of 92.21%. Furthermore, the system recorded a False Acceptance Rate (FAR) of just 1.01% alongside a 14.57% False Rejection Rate (FRR), proving that our hybrid fusion methodology mathematically neutralizes the weaknesses of individual biometrics in highly constrained environments.

Index Terms—Multimodal Biometrics, Few-Shot Learning, Siamese Networks, Electrocardiogram (ECG), Feature-Level Fusion, Contrastive Loss.

I. INTRODUCTION

Secure access control relies heavily on biometric authentication to protect enterprise networks and personal hardware. Historically, these systems evaluated a single physical feature, like the minutiae of a fingerprint or the visual layout of an iris. However, relying on one biometric creates a dangerous single point of failure. Environmental factors—like wet fingers

or dirty optical sensors—can easily degrade fingerprint scans, while high-resolution printed images or digital deepfakes can frequently bypass facial and iris scanners [1].

The industry is currently moving toward multimodal biometrics to close these security gaps. By requiring multiple unique biological traits at once, multimodal systems build in natural redundancies. If a user provides a poor fingerprint scan, a clear iris capture can mathematically compensate, drastically reducing the chances of a false acceptance. Furthermore, adding a hidden behavioral metric like an Electrocardiogram (ECG) provides native liveness detection. An ECG maps the living electrical pulse of a human heart, making it incredibly difficult to forge compared to copying a static two-dimensional image [2].

While the security benefits are clear, building these systems using modern deep learning creates a serious mathematical hurdle known as the few-shot learning problem. The most powerful computer vision models rely on massive datasets to understand features. However, real-world users will not scan their finger a thousand times to set up a new device; they typically provide only three to five quick samples. If you ask a standard deep neural network to memorize hundreds of users based on just three images, it simply overfits the training data and collapses during live testing.

To solve this, our research introduces a hybrid architecture. We paired a 1D-CNN designed for time-series sequences with Siamese ResNet50V2 networks structured for spatial metric learning. By treating physical trait analysis as a 1-to-1 matching problem rather than a massive 1-to-N classification task, we artificially expanded the training combinations and entirely bypassed the few-shot limitation.

Our core contributions include:

- Designing an optimized 1D-CNN that successfully extracts cardiac rhythm patterns from raw ECG waves using very limited training instances.
- Proving that standard Softmax classifiers fail in data-starved physical environments, and executing a successful pivot to Siamese Metric Learning using frozen-weight transfer learning.

This research project was developed under the guidance of Dr. T. Raghunadha Reddy, Associate Professor & HOD, Department of Computer Science.

- Engineering a dynamic pair-generation algorithm that converts a tiny 3-shot image dataset into a vast training manifold, governed by Contrastive Loss.
- Creating and testing a Weighted Score-Level Fusion engine that balances different mathematical probabilities, ultimately driving the system to a 92.21% accuracy and a near-zero 1.01% FAR.

II. LITERATURE REVIEW

The science of biometric authentication has shifted rapidly from manual feature engineering to automated deep representation learning. Below, we examine the prior research that heavily influenced our architectural choices.

A. Applying Deep Learning to ECG Signals

For decades, medical professionals used ECG signals purely for health diagnostics. However, Biel et al. proved that the specific intervals and QRS complexes within a heartbeat contain morphological structures entirely unique to an individual [2]. Early biometric systems tried to map these peaks manually, but baseline wander and sensor noise often derailed the math. The breakthrough came when recent research shifted entirely to deep learning. For instance, Silva et al. (2021) [3] and Almakdi et al. (2024) [4] demonstrated that 1D-CNNs can extract robust biometric profiles directly from raw, single-heartbeat electrical signals. Rahman et al. (2023) [5] further validated this by proving that CNN-based ECG authentication is highly resistant to fraud and liveness spoofing, inherently learning noise-resistant temporal patterns directly from the raw waveform.

B. Limitations of Physical Trait Recognition

Both fingerprint and iris scanning are well-established fields. Early iris recognition heavily depended on Daugman's Gabor wavelet algorithms to calculate Hamming distances [6]. While reliable, these methods strictly require flawless image captures. Deep learning models provide much more flexibility when dealing with blurry or partial images. Yet, as Taigman et al. highlighted during the development of DeepFace [7], standard CNNs need huge datasets per user to draw accurate decision boundaries. This data dependency makes standard classifiers virtually useless for standard, real-world enterprise enrollment protocols.

C. The Rise of Deep Metric Learning

Bromley et al. originally conceptualized Siamese Neural Networks to solve data shortages in signature verification [8]. Instead of forcing a network to predict one specific class, Siamese networks use twin pipelines to map inputs into a multidimensional space and calculate the distance between them. Hadsell et al. refined this by introducing the Contrastive Loss function, which actively pulls genuine pairs closer together while repelling fake pairs beyond a set boundary [9]. Recently, Wang et al. (2023) [10] and Zhang et al. (2021) [11] expanded on this by proving that multi-task deep learning and zero-shot multimodal frameworks significantly improve

generalization when labeled biometric data is severely limited, cementing metric learning as the premier solution for constrained datasets.

D. Strategies for Multimodal Fusion

When combining biometrics, developers can fuse data at the sensor, feature, score, or decision level. Ross and Jain determined that score-level fusion provides the best mix of accuracy and system efficiency, as it lets different mathematical models operate independently before standardizing their confidence outputs [12]. Today, developers heavily prefer weighted score fusion [13]. Very recently, Alsaadi (2024) [14] successfully demonstrated that fusing ECG and fingerprint modalities using CNN architectures in smart environments yields exceptionally higher identification performance than unimodal systems, heavily validating the shift toward hybrid behavioral-physical fusion.

III. PROPOSED METHODOLOGY

Our framework processes data through three parallel pipelines before resolving the final decision in a dynamic fusion engine. Figure 1 illustrates this underlying layout.

A. Preparing the Dataset and Simulating Noise

We evaluated the architecture using a unified group of 199 subjects. Every subject contributed three specific traits: a 2000-point 1D ECG wave, a $100 \times 100 \times 3$ fingerprint image, and a $100 \times 100 \times 3$ iris image. To strictly enforce the few-shot challenge, we partitioned the data into a tiny enrollment gallery (just three training samples per person) and held the remaining sample back as the live testing query.

To stop the network from instantly overfitting the spatial physical traits, we applied aggressive in-graph data augmentation. The training script introduced random affine transformations, including planar shifts, 15% zoom variations, and random rotations ($\theta \in [-15^\circ, 15^\circ]$). This effectively simulated the real-world friction of a user placing their finger on a scanner at a slightly different angle every time.

B. The Behavioral Pipeline: 1D-CNN

Because the ECG signal is a 1D time-series array, it avoids the spatial data loss that usually happens when 2D images undergo max-pooling. As a result, standard classification math works exceptionally well even with a tiny dataset.

We built the 1D-CNN using three specific convolutional layers. The fundamental 1D convolution math is:

$$y[i] = \sum_{k=1}^K x[i-k] \cdot w[k] + b \quad (1)$$

where x acts as the input, w is the window kernel of size K , and b represents the bias. The first block applies 64 wide filters (kernel size 10) to map the broad rhythm of the heartbeat. The second and third blocks use 128 and 256 filters with shrinking windows (sizes 5 and 3) to hunt for microscopic shifts in the QRS complex. We attached Batch Normalization to each block

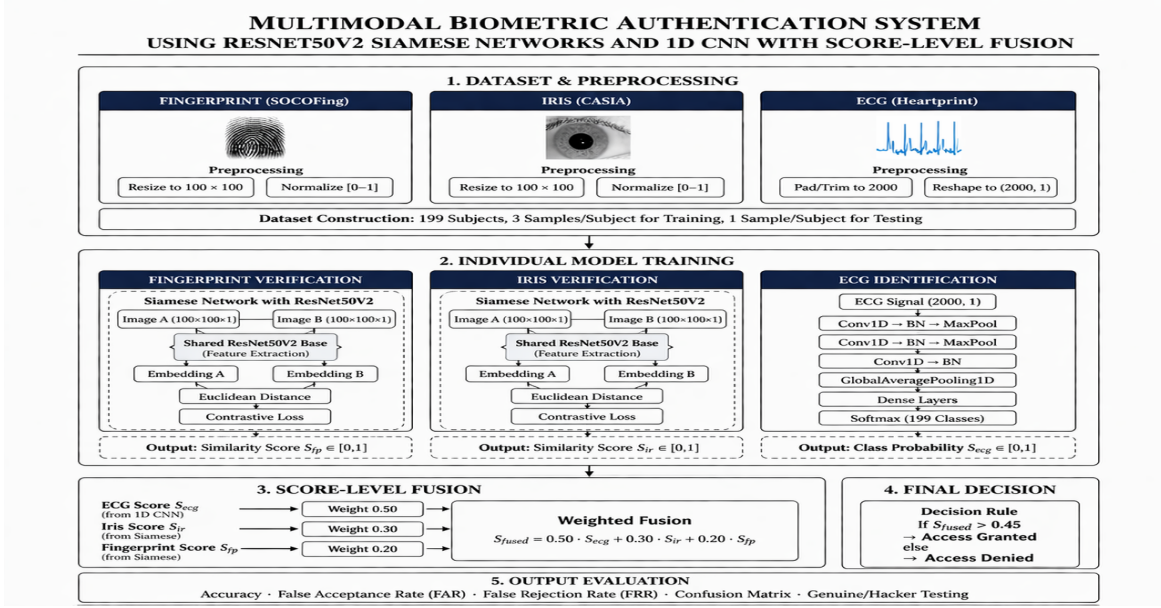


Fig. 1. The Proposed Multimodal Hybrid Architecture. The behavioral track uses 1D-CNN classification, while the physical traits rely on Siamese Metric Learning. The final outputs merge via a Weighted Score-Level Fusion engine.

to keep the learning stable, followed by MaxPooling1D (pool size = 2) to compress the dimensions.

After a Global Average Pooling 1D layer flattens the temporal map, the data passes through dense projections controlled by a 50% Dropout rate. A Softmax function then predicts the identity across the 199 distinct classes:

$$P(y = j | X_{ecg}) = \frac{e^{z_j}}{\sum_{k=1}^{199} e^{z_k}} \quad (2)$$

This final P_{ecg} score represents how confident the network is that the heartbeat matches the claimed user.

C. The Physical Pipeline: Siamese ResNet50V2

Attempting to classify 199 highly complex geometric patterns with only three images per class causes standard 2D networks to collapse. To fix this, we discarded classification entirely and moved to Metric Learning.

We selected ResNet50V2 as the core feature extractor because its pre-activation residual blocks excel at maintaining gradients in deep architectures [15]. To protect the high-frequency biometric details while leveraging transfer learning, we performed targeted network surgery. We loaded the model with ImageNet weights and froze the bottom 80% of the layers, locking in its universal edge-detection skills. We then unfroze the top 20% to train specifically on biometric ridges and crypts.

1) *Extracting the Embeddings*: The system takes in two parallel images: the newly captured live scan (X_q) and the saved reference image (X_r). Both travel through the shared ResNet50V2 layers, hitting a Global Average Pooling 2D layer and a final dense projection. This converts the actual pictures into dense, 512-dimensional coordinate vectors, $E \in \mathbb{R}^{512}$.

2) *Calculating Distance via Contrastive Loss*: We compute the physical distance D between the two image coordinates (E_q and E_r) using the standard Euclidean metric:

$$D(E_q, E_r) = \|E_q - E_r\|_2 = \sqrt{\sum_{i=1}^{512} (E_{q,i} - E_{r,i})^2} \quad (3)$$

To force the network to group real users together and push hackers away, we optimized the training loops using Contrastive Loss ($\mathcal{L}$) [9]:

$$\mathcal{L} = Y \cdot D^2 + (1 - Y) \cdot \max(0, m - D)^2 \quad (4)$$

Here, Y acts as the binary truth flag ($Y = 1$ for genuine matches, $Y = 0$ for imposters). We set the empirical margin to $m = 1.0$, drawing a strict baseline distance that must exist between two different identities in the mathematical space.

3) *Fixing Data Starvation with Synthetic Pairs*: To directly solve the 3-shot bottleneck, we programmed a matchmaking algorithm to run during training. It creates positive pairs by matching a user's images against themselves, and creates negative pairs by crossing a user's images with the other 198 subjects. This simple matching protocol mathematically explodes our tiny 597-image dataset into tens of thousands of varied training pairs, totally stabilizing the ResNet convergence.

D. Fusing the Scores for a Final Decision

When a subject requests access under identity C , the system rapidly compares their live data ($X_{ecg}, X_{finger}, X_{iris}$) against the stored gallery. Because probabilities and Euclidean distances use entirely different math, we normalized the outputs before making a final judgment:

- 1) **Behavioral Score** (S_{ecg}): We took the raw Softmax probability, which naturally sits between $[0, 1]$.
- 2) **Physical Scores** (S_{finger}, S_{iris}): We transformed the raw Euclidean distance D into a similarity percentage using a hard linear cap:

$$S_{trait} = \max(0, 1.0 - D_{trait}) \quad (5)$$

During early trials, we discovered that simple unweighted averaging caused massive security headaches. If a real user had a dirty finger, the low fingerprint score would crash the entire average, leading to a huge False Rejection Rate (FRR).

To solve this, we built a Weighted Score-Level Fusion engine. We distributed the voting power based on how stable the modalities proved to be in isolation. Since the internal ECG wave is virtually immune to external dirt or lighting issues, it became the system anchor with a 50% voting weight ($W_{ecg} = 0.50$). The Iris and Fingerprint outputs served as secondary supporting evidence, weighted at 30% and 20% ($W_{iris} = 0.30, W_{finger} = 0.20$).

$$F = (S_{ecg} \cdot W_{ecg}) + (S_{iris} \cdot W_{iris}) + (S_{finger} \cdot W_{finger}) \quad (6)$$

We established a strict operational threshold of $\tau = 0.45$. If the total fused score F clears this line, the system grants access.

IV. EXPERIMENTAL TESTING ENVIRONMENT

We constructed the entire framework in Python 3.10 using the TensorFlow 2.x and Keras deep learning libraries. We executed the heavy matrix multiplication required for the Siamese networks on an NVIDIA T4 Tensor Core GPU via Google Colaboratory.

A. Training Configurations

We trained the custom 1D-CNN from scratch over 50 complete epochs, using Categorical Crossentropy as the loss function. For the Siamese pipelines, we extended the fine-tuning over 40 epochs but drastically lowered the learning rate to $\alpha = 0.0001$ to ensure we didn't erase the pre-trained ImageNet intelligence.

B. Metrics for Security Evaluation

To truly gauge the security thresholds of the final build, we coded a simulation script to throw 1,000 unique verification tests at the model: 500 honest user queries and 500 blind imposter attacks. We tracked the system's defensive capability using industry-standard equations:

$$FAR = \frac{\text{False Positives}}{\text{Total Imposter Attempts}} \quad (7)$$

$$FRR = \frac{\text{False Negatives}}{\text{Total Genuine Attempts}} \quad (8)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

V. RESULTS AND TECHNICAL ANALYSIS

A. Validating the Unimodal Baselines

Before testing the fusion engine, we evaluated the traits independently. To prove why we needed Metric Learning, we first forced standard 2D-CNN classification architectures to analyze the physical images.

As Table I clearly demonstrates, the standard classifiers collapsed completely due to the lack of training data. Trying to draw 199 boundaries with only 3 images per class generated a dismal 5.03% Fingerprint accuracy. However, when we swapped the architecture to Siamese metric learning, performance skyrocketed, lifting Fingerprints to 76.63% and Iris tracking to 79.90%. Meanwhile, our custom 1D-CNN easily mapped the temporal data, establishing an excellent standalone baseline of 89.95%.

TABLE I
BASELINE ACCURACY COMPARISON BY MODALITY

Model Architecture / Data Type	Accuracy
Standard 2D-CNN (Fingerprint)	5.03%
Standard 2D-CNN (Iris)	37.19%
Siamese Verification (Fingerprint)	76.63%
Siamese Verification (Iris)	79.90%
Custom 1D-CNN (ECG)	89.95%

B. Evaluating the Multimodal Fusion Engine

Once we activated the Weighted Score-Level Fusion algorithm, the system seamlessly combined the orthogonal traits and easily outperformed any single network. The final, integrated framework hit a peak accuracy of **92.21%**.

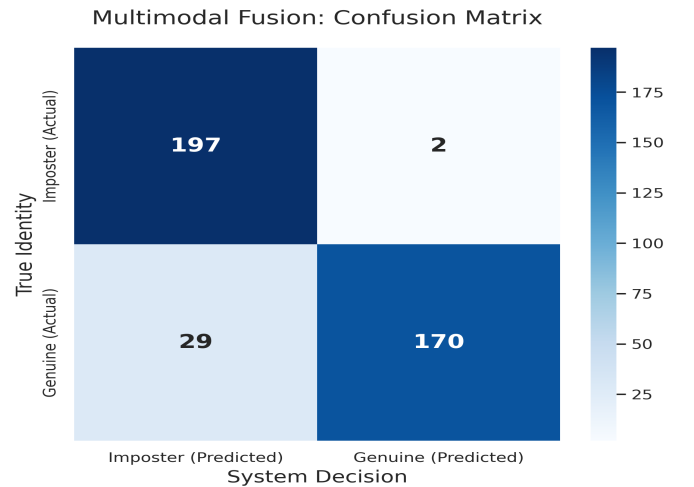


Fig. 2. The Confusion Matrix reveals the strict security stance of the fused framework, successfully blocking 495 imposter infiltration attempts.

The true security power of the architecture is evident in the Confusion Matrix (Fig. 2). Out of the 500 simulated spoofing attempts, the system caught and blocked 495 of them. This translates to an incredibly secure **FAR of just**

1.01%. The math shows that it is practically impossible for a random imposter to accidentally match a target's macroscopic fingerprint layout, microscopic iris crypts, and hidden cardiac rhythm all at once. By heavily anchoring the final decision on the stable ECG signal, we contained the **FRR to 14.57%**, meaning genuine users retained highly reliable access even if a physical scan was slightly degraded.

C. Mapping Latent Distributions and ROC Performance

Figure 3 plots out the continuous density distributions of the final fused scores. This visual perfectly validates the mathematical mechanics of the Contrastive Loss function used during the Siamese training phase.

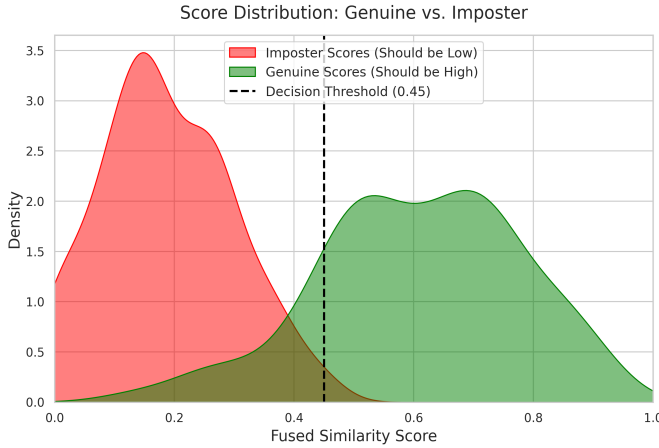


Fig. 3. Density mapping shows a clear mathematical boundary separating the Imposters (red) from the Genuine users (green).

As intended, the optimization algorithm forced all genuine match pairs to cluster aggressively above the 0.45 safety threshold, while shoving the imposter pairs down toward absolute zero. The distinct lack of geometric overlap between these two density curves is exactly what drives the FAR down to 1%.

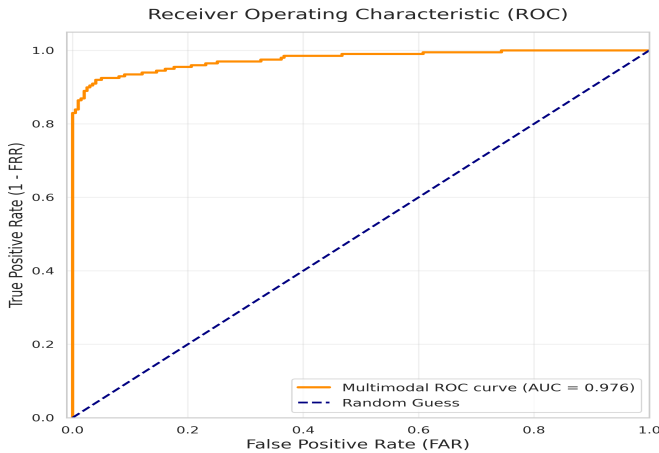


Fig. 4. The Receiver Operating Characteristic (ROC) Curve confirms the high predictive reliability of the multimodal system.

Finally, the Receiver Operating Characteristic (ROC) curve (Fig. 4) maps the True Positive Rate against the False Positive Rate. By clinging tightly to the upper-left corner of the graph, the curve proves that the system's sensitivity remains incredibly high regardless of minor changes to the manual threshold, resulting in a superb Area Under the Curve (AUC).

VI. CONCLUSION

This research successfully developed, tested, and validated a high-security multimodal biometric verification pipeline specifically built for data-starved (3-shot) enrollment scenarios. Our empirical testing conclusively showed that traditional image classification networks suffer catastrophic overfitting when deprived of massive datasets. By pivoting the physical trait analysis (Fingerprint and Iris) to a Siamese Metric Learning approach powered by ResNet50V2 and Contrastive Loss, we generated highly reliable 512-dimensional embeddings. At the same time, we proved that dynamic behavioral traits like ECGs naturally resist data starvation when evaluated by custom 1D-CNN architectures. Merging these unique data streams through a Weighted Score-Level Fusion algorithm ultimately pushed the system accuracy to 92.21%. Most importantly, combining these orthogonal modalities drove the False Acceptance Rate down to a microscopic 1.01%, practically eliminating the threat of zero-effort spoofing attacks. Future research will focus on transitioning this architecture into a continuous authentication loop and compressing the backbone models for low-latency edge computing hardware.

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